

Chronic Kidney Disease



CHRONIC KIDNEY DISEASE

INTRODUCTION



- CKD is a worldwide public health problem.
- A marked increase in the incidence and prevalence of end stage renal disease (ESRD) since the mid-1980's.
- In 1990 426,000 patients on dialysis globally, 1.49 million in 2000 and a projected 2.5 million in 2010.
- An estimated 500 million suffer from CKD.
- In (1999-2000), 9.63% of the US pop'n has CKD.

CONT...

- CKD is associated with significant morbidity & mortality and a reduced quality of life.
- CKD IS UNDER DIAGNOSED AND UNDERTREATED BUT IS POTENTIALLY PREVENTABLE.
- CKD can be detected using simple lab tests and treatment can prevent or delay complications of decreased kidney function, slow the progression of kidney disease and reduce the risk of cardiovascular disease (CVD).

CHRONIC KIDNEY DISEASE

DEFINITION AND CLASSIFICATION

- The presence of markers of kidney damage for 3 months, as defined by structural or functional abnormalities of the kidney with or without decreased glomerular filtration rate (GFR), manifest by either pathological abnormalities or other markers of kidney damage, including abnormalities in the composition of blood or urine, or abnormalities in imaging tests.
- OR The presence of $\text{GFR} < 60 \text{ mL/min/1.73 m}^2$ for 3 months, with or without other signs of kidney damage as described above.

DEFINITION AND CLASSIFICATION OF CKD

STAGE	DESCRIPTION	GFR (ML/ MIN/ 1.73M ²)	TREATMENT
1	KIDNEY DAMAGE WITH NORMAL OR ↑ GFR	≥ 90	T IF TRANSPLANT RECIPIENT
2	KIDNEY DAMAGE WITH MILD ↓ GFR	60- 89	
3	MODERATE ↓ GFR	30- 59	
4	SEVERE ↓ GFR	15- 29	
5	KIDNEY FAILURE	< 15 (OR DIALYSIS)	D IF DIALYSIS (HD OR PD)

CHRONIC KIDNEY DISEASE :EVALUATION

- GFR is the single most useful qualitative index of kidney function in health and disease. The gold standard for measurement of GFR is inulin clearance. Radionuclides and contrast agents(e.g. iothalamate) can be used to measure GFR.
- In everyday practice serum creatinine is used as a surrogate for GFR but it is a gross estimate of GFR.
- Prediction equations that take into account creatinine levels, age, sex, body size and race are better measures of GFR.
- The Cockcroft-Gault formula simple to use:

$$\text{CrCl(ml/min)} = (140 - \text{age}) \times \text{body wt. in Kg} / \text{Scr in mg/dl} \times 72$$

Multiply by 0.85 in females

The Modification of Diet in Renal Disease(MDRD) equation is even better and takes additional parameters including race into account but requires a calculator.

$$\text{GFR, in mL/min per 1.73 m}^2 = 186.3 \times \text{SCr} (\exp[-1.154]) \times \text{Age} (\exp[-0.203]) \times (0.742 \text{ if female}) \times (1.21 \text{ if black}) \text{ where exp is the exponential.}$$

CKD: PATHOPHYSIOLOGY OF PROGRESSION



- In 1982 Brenner and colleagues put forward their hypothesis on the mechanism of progression of CKD.
- In response to nephron loss compensatory glomerular hemodynamic changes occur which eventually lead to progressive loss of kidney function.
- With renal mass reduction the remaining nephrons undergo hypertrophy with reduced arteriolar resistance and increased glomerular blood flow. As a consequence glomerular hypertension and $\uparrow$ GFR(glomerular hyperfiltration) occur.
- Angiotensin II plays a central role in this process.

CKD: PATHOPHYSIOLOGY OF PROGRESSION



- Angiotensin II preferentially increases the tone of the efferent arteriole hence increasing glomerular hydraulic pressure and the filtration fraction. The increase in glom capillary pressure increases the filtration of macromolecules.
- AG II also alters glomerular perm selectivity by mediating contraction of the glomerular foot processes.
- AG II modulates renal cell growth which in turn may contribute to tubulointestinal injury.
- AG II also contributes to accumulation of extracellular matrix through stimulating production of PAI-1.
- AF II may enhance the inflammatory process by directly stimulating macrophage activation and phagocytosis.

CKD: PATHOPHYSIOLOGY OF PROGRESSION



- Proteinuria which is consequence of glomerular hyperfiltration and is a marker of glomerular injury also contributes to progressive nephron damage.
- Reabsorption of the filtered protein by the proximal tubule activates inflammatory responses (complement, cytokines, growth factors and vasoactive substances) which eventually causes tubulointerstitial injury (tubular dilatation, interstitial fibrosis).
- Clinical studies have shown that the protein excretion rate is the single best predictor of progression and ESRD.
- Therapy to reduce proteinuria is associated with slowing of progression of CKD.

CKD: PATHOPHYSIOLOGY OF PROGRESSION



- Risk factors associated with progression include: Proteinuria $> 1.5\text{gm}/24\text{hrs}$ or protein to creatinine excretion > 1 , Hypertension, African American race, male sex, obesity, diabetes melitus, hyperlipidemia, smoking, high-protein diet, phosphate retention and metabolic acidosis.

CKD: PATHOPHYSIOLOGY OF PROGRESSION



- Renal injury → Reduction in renal mass → Glomerular capillary hypertension → ↑glomerular permeability to macromolecules → ↑filtration of plasma proteins(proteinuria) → Excessive tubular protein reabsorption → Tubulointerstitial inflammation → Renal scarring

CKD: CLINICAL MANIFESTATIONS AND COMPLICATIONS



- Manifestations dependent upon severity of renal impairment and underlying disease.
- Most patients with early(Stage 1&2) CKD asymptomatic.
- Polyuria/nocturia may be early manifestations.
- In late CKD uremic manifestations like easy fatigability, anorexia, vomiting, deficits in cognitive function, etc will be present.
- Complications like anemia, electrolyte or acid base disorder, renal osteodystrophy may be the manifestations patients with advanced CKD may come with.

CKD :SCREENING



- All individuals during routine health encounters.
- Targeted evaluation of those at increased risk : those with diabetes, hypertension, CVD, family history of diabetes/hypertension or CKD, etc
- Urinary exam for protein/albumin, creatinine assessment ± additional tests
- Once diagnosis is established identify and treat cause & potentially reversible factors and classify Stage of CKD.

CKD:MANAGEMENT



- Establish the cause of CKD and treat (if possible)
- Identify and manage unexpected (acute) deterioration in kidney function.
- Slow progression.
- Manage complications of CKD.Special emphasis on CVD.
- Prepare for Renal Replacement Therapy.

CKD: CAUSES



- Diabetes : the most common cause of ESRD in the Western World
- Hypertension : a common cause of ESRD among African Americans
- Chronic glomerulonephritis
- Cystic kidney diseases: Autosomal Dominant Polycystic Kidney Disease(ADPKD) the commonest.
- Structural anomalies of the kidneys and urinary tract: important causes of ESRD among children
- Chronic tubulointerstitial nephritis

CKD: CAUSE IDENTIFICATION



- Many good reasons to identify cause:
- i) to arrest the process and prevent further renal loss, when possible
- ii) to provide the best advice on prognosis
- iii) to give genetic counseling where appropriate
- iv) to inform the decision on eligibility for renal transplantation
- Establishing cause usually simple: history, past medical records.

CKD:

REVERSIBLE CAUSES OF RENAL DYSFUNCTION



- Identify and treat causes of reversible renal dysfunction.
- I) Decreased renal perfusion: hypovolemia, hypotension(e.g. Myocardial dysfunction), sepsis, drugs that ↓GFR
- II) Nephrotoxic Drugs
- III) Urinary Tract Obstruction
- IV) Others: Disease relapse, hypertension, hypercalcemia

CKD MANAGEMENT: SLOWING PROGRESSION



- Focus on modifiable risk factors
 - I) Dietary interventions: Protein restriction, low phosphate diet, manipulation of lipids including fish oil supplements, salt restriction
 - II) Pharmacologic interventions: BP control & proteinuria reduction- ACE-I/ARB $\pm$ other antihypertensives. ? Lipid lowering agents
 - III) Other interventions : Cessation of Smoking, restricting alcohol consumption

Treatment of Complications of Renal Dysfunction



- A) HYPERTENSION: Present in $\geq 80\%$ of those with advanced CKD.
- BP Targets: $<130/80$ (MAP 92) if proteinuria $< 1\text{Gm}/24$ hours, $<125/75$ (MAP 90) if proteinuria $> 1\text{Gm}/24$ hrs
ACE-I, Salt restriction/diuretic, ARB, Non-dihydropyridine Ca channel blockers, alpha blockers, etc

CKD Management

Treatment of Complications of Renal Dysfunction

- B) ANEMIA: Normocytic-normochromic & primarily due to EPO deficiency, stabilises at ~Hct 25% if untreated.
 - associated with increased morbidity & mortality. Is an important & independent risk factor for the development and progression of LVH and CHF in CKD as well as for increased CV mortality
 - evaluation to begin when Hgb < 12gm/dl in females & < 13.5 gm/dl in males.
 - Diagnosis of anemia of CKD made after excluding non-renal causes in a person with ↓GFR

CKD Management

Treatment of Complications of Renal Dysfunction



- B) Anemia: evaluation to include RBC indices, reticulocyte count, Ferritin, %transferrin sat.(iron/tibc $\times$ 100), stool occult blood
- Treatment target: Hemoglobin 11- 12gm/dl
 - Iron : oral or IV to maintain % transferrin saturation $>20\%$ ($<50\%$) and Ferritin $>100\text{ng/ml}$ (500ng/ml)

CKD Management

Treatment of Complications of Renal Dysfunction



- C) Hyperphosphatemia: Phosphate retention from early CKD but normal balance maintained until GFR~30ml/min
(but at the cost of developing secondary hyperparathyroidism & renal osteodystrophy).
- Hyperphosphatemia is associated with vascular calcifications and increased mortality.

CKD Management

Treatment of Complications of Renal Dysfunction



- C)Hyperphosphatemia: -Treatment includes dietary phosphate restriction and the use of oral phosphate binders.
- ↓dietary phosphate to 800- 1000mg/dl.
- Phosphate binders include Calcium containing binders(Ca carbonate or acetate),Sevelamer and Lanthanum Carbonate.

CKD Management

Treatment of Complications of Renal Dysfunction



- ▶ D) Renal Osteodystrophy: The principal types of renal bone disease include osteitis fibrosa, osteomalacia and adynamic bone disease.
- ▶ Osteitis fibrosa results from secondary hyperparathyroidism- $\uparrow$ PTH when $GFR < 40-70 \text{ ml/minute}$. Serial measurements of plasma PTH, serum Ca and PO_4 , $Ca \times PO_4$ ($\pm 1,25$ Vitamin D levels) needed for adjusting treatment.

Treatment of Complications of Renal Dysfunction



- ▶ IV) Renal Osteodystrophy: Treatment includes dietary PO₄ restriction, PO₄ binders, active Vitamin D sterols (Calcitriol, α calcidol, paricalcitol, doxercalciferol, etc)
 - Dose adjustment according to severity to avoid risk of developing hypercalcemia, hyperphosphatemia or over suppression of PTH with adynamic bone disease.
 - Calcimimetics: a new class of drugs that ↑ the sensitivity of the Calcium Sensing Receptor to Ca. Complementary & synergistic suppression of PTH with Vitamin D analogues. CINACALCET is the prototype. Approved for use in dialysis patients only.
 - Parathyroidectomy for limited indications

CKD Management

Treatment of Complications of Renal Dysfunction



- ▶ E) Cardiovascular Disease (CVD):CKD is an independent & important risk factor for CVD.
- ▶ The risk of death from CVD(CAD,CHF) is higher than the risk of eventually requiring RRT.
- ▶ Traditional CV risk factors :HTN usually with LVH, DM, dyslipidemia, smoking, old age are more prevalent in CKD populations.
- ▶ Non-traditional risk factors unique to CKD include: uremic toxins, abnormal bone mineral metabolism, anemia, proteinuria, hyperhomocysteinemia and an increased inflammatory-poor nutrition state.
- ▶ Renal impairment probably sensitizes cells to the injurious effects of HTN, hyperglycemia and hyperlipidemia.

CKD Management

Treatment of Complications of Renal Dysfunction



- ▶ E) Cardiovascular Disease : CKD can be considered a coronary heart disease risk equivalent.
- ▶ Secondary prevention measures: mostly similar to recommendations for those with CVD risk factors without renal dysfunction. Individual decisions for ASA b/c of ↑risk of bleeding.
- ▶ Measures aimed at correcting non-traditional risk factors may improve CV outcome in CKD.

CKD Management

Treatment of Complications of Renal Dysfunction



- F) Other complications: Volume overload, metabolic acidosis, hyperkalemia are issues to be addressed.
- Malnutrition: common in advanced CKD due to ↓food intake, ↓intestinal absorption/digestion and metabolic acidosis. Serially monitor weight & serum albumin. Involve a dietician if possible. Avoid over restriction of proteins.

CKD Management:

Preparation for & Initiation of RRT



- Early identification of those with CKD and referral to a specialist(nephrologist) may contribute to improvement in outcome.
- Choice of Renal Replacement Therapy: Counsel patient about Hemodialysis, Peritoneal Dialysis and renal transplantation.
- Transplantation is the treatment of choice for most patients with ESRD.
- Prepare for the chosen RRT modality.